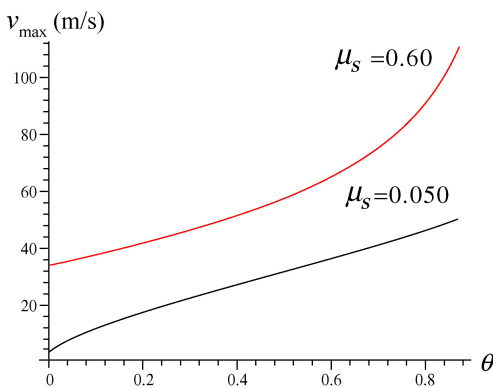

Additional Problems for Ch 6 HRW – Forces 2

Addisionele Probleme vir Hfst 6 HRW – Kragte 2

	9 th Edition	8 th Edition
Questions	1, 6, 7, 9	3, 6, 7, 11
Problems	7, 11, 23, 25, 28, 57, 63, 68, 77, 80, -	3, 13, 25, 23, 30, 55, 79, 62, 73, 76, -

7	3	(a) $1.6 \times 10^2 \text{ N}$ (b) 1.8 m/s^2	(a) $1.9 \times 10^2 \text{ N}$ (b) 0.56 m/s^2
11	13	(a) $3.0 \times 10^2 \text{ N}$ (b) 1.3 m/s^2	(a) $3.0 \times 10^2 \text{ N}$ (b) 1.3 m/s^2
23	25	0.37	
25	23	$1.0 \times 10^2 \text{ N}$	
28	30	4.9 kg	3.3 kg
57	55	1.81 m/s	
63	79	(b) 234 N (c) 0.59	(b) 240 N (c) 0.60
68	62	(a) $v_{\max} = \sqrt{Rg \tan(\theta + \tan^{-1} \mu_s)} = \sqrt{\frac{Rg(\tan \theta + \mu_s)}{1 - \mu_s \tan \theta}}$ (b)  (c) $v = 46.1 \text{ m/s} = 166 \text{ km/h}$ when $\theta = 10^\circ = 0.175 \text{ radian}$. (d) $v = 21.2 \text{ m/s} = 76.2 \text{ km/h}$ (c) $v = 41.3 \text{ m/s} = 149 \text{ km/h}$ when $\theta = 10^\circ = 0.175 \text{ radian}$. (d) $v = 21.2 \text{ m/s} = 76.2 \text{ km/h}$	
77	73	147 m/s	
80	76	$6.2 \times 10^3 \text{ N}$	